MITTAG-LEFFLER COHOMOLOGY SYSTEMS FOR PSEUDO-COHERENT COMPLEXES MODULO POWERS OF AN IDEAL

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Allowed standard prerequisites (editorial).

- For a bounded above complex of finite free modules the systems of cohomology of the reductions modulo I^n satisfy Mittag-Leffler and the I-adic completion of the cohomology equals the limit of the cohomologies of the reductions. invoked at proof lines 29010-29011. Gives the final identification.

Source reference. Original lemma 0EGU: sources/stacks-more-algebra/pinned-original-source.tex, source lines 28867–28867. Cited at proof lines 29010-29010 for the Mittag-Leffler property; the conclusion is used inside the counted span.

Source reference. Original lemma 0EGS: sources/stacks-more-algebra/pinned-original-source.tex, source lines 28523–28523. Cited at proof lines 29010-29011 for the final limit identification; the conclusion is used inside the counted span.

Lemma 0.2. *Let I be an ideal of a Noetherian ring A. Let $K \in D(A)$ be pseudo-coherent. Set $K_n = K \otimes^{\mathbf{L}}_A A/I^n$. Then for all $i \in \mathbf{Z}$ the system $H^i(K_n)$ satisfies Mittag-Leffler and $\lim H^i(K)/I^n H^i(K)$ is equal to $\lim H^i(K_n)$.*

Proof. We may represent K by a bounded above complex $P^\bullet$ of finite free A-modules. Then K_n is represented by $P^\bullet/I^n P^\bullet$. Hence the Mittag-Leffler property by Lemma 0EGU. The final statement follows then from Lemma 0EGS. □

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